MIL-R-51397C(EA)
30 March 1981
SUPERSEDING
MIL-R-51397B(EA)
31 October 1979

MILITARY SPECIFICATION

REFILL KIT, CHEMICAL AGENT AUTOMATIC ALARM: ABCA-M229

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes requirements for one type of refill kit.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-D-51388 - Detector Cell, Chemical Agent.

FSC 6665

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-650	-	Explosive: Sampling, Inspection and Testing.
MIL-STD-810	-	Environmental Test Methods.
MIL-STD-1188	-	Commercial Packaging of Supplies and Equipment.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL 5-15-4700	-	Refill Kit, Chemical Agent Automatic Alarm: ABCA-M229.
C5-15-6569	-	Packaging Refill Kit, ABCA-M229.
D136-40-79-2	-	Tester, Agent Conversion and Detection, Q197A1.
D136-41-2136	-	Tester, Leakage and Resistance, Filter Assembly, Q198.
E136-41-1520	-	Portable Gas, Generator, Q219.

PUBLICATIONS

DEPARTMENT OF THE ARMY TECHNICAL MANUAL

TM3-6665-225-12	-	Alarm, Chemical Agent, Automatic: Portable, ABCA-M8.

CHEMICAL SYSTEMS LABORATORY PRODUCT ASSURANCE DIRECTORATE

136-305-1	-	Procedure for Chemical Analysis of Chemical Agents Tabor (GA) and Sarin (Gb).
136-305-16	-	Procedure for Chemical Analysis of Methane Sulfonyl Fluoride (MSF).
136-305-17	-	Procedure for Chemical Analysis of o,s-Diethyl Methyl Phosphonothialate (OSDMP).

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

E148 - Apparatus for Microdetermination of Nitrogen by the Dumas Method, Specification for.
E191 - Apparatus for Microdetermination of Carbon and Hydrogen in Organic and Organo-Metallic Compounds, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-15-4700 and on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings and in this specification.

3.1.2 Components. All components of the refill kit shall conform to the specifications and drawings listed on TDPL 5-15-4700 and subsidiary drawings.

3.2 Manufacture and assembly. The refill kit shall be manufactured and assembled as specified on Drawing D5-15-4700.

3.3 Performance.

3.3.1 Isonitrosobenzoylacetone (1-phenyl -1, 2, 3-butanetrione 2-oxime), (IBA) melting point and composition (see 6.3). The IBA shall be a white granular solid with a melting point between 129°C and 134°C, but over no more than a 2°C increment within this range. The chemical composition of the IBA shall conform with the requirements of table I. Water content of the IBA shall not exceed one percent by weight. The particle size of the IBA shall be such that a minimum of 90 percent of a given weight will pass through a 425 micrometer (United States Standard No. 40) sieve. The shaking or vibrating technique of sieving shall be employed; the IBA shall not be rubbed through the sieve. Test as specified in 4.4.4.1. Note: Nonmetallic or stainless steel sieves should be used. IBA that has been in contact with metal sieves, other than those fabricated from stainless steel, shall not be returned to the lot.

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TABLE I. Chemical composition of IBA

Elemental analysis	Percent by weight	
	Minimum	Maximum
Carbon	62.55	63.15
Hydrogen	4.41	5.01
Nitrogen	7.03	7.63

3.3.2 Benzenesulfonyl chloride concentration of bottled simulant solution. The concentration of benzenesulfonyl chloride of the bottled simulant solution shall be as specified by Drawing C5-15-5505 when tested as specified in 4.4.4.2.

3.3.3 Chemical reservoir.

3.3.3.1 Visual examination. The IBA powder packet (Drawing C5-15-5599) shall be installed with the backing-paper side toward the cutter-plunger in the cap. There shall be no contaminants or particles immersed in or floating upon the buffer solution discernible to the unaided eye when examined visually in accordance with 4.4.4.3.1.

3.3.3.2 Cutter-plunger operation. Operating the cutter-plunger and shaking the reservoir for a maximum of 3 minutes shall result in the IBA powder being released from the packet. The presence of pieces of the IBA powder packet or the cutter-plunger in the solution shall not be a cause for rejection when tested as specified in 4.4.4.3.2.

3.3.3.3 Use test with detector unit. The electrolyte resulting from the mixing of the IBA and the liquid within the chemical reservoir shall be capable of successfully operating a detector unit for a period of at least 12 hours (test must be completed within 24 hours after mixing the IBA and the liquid). At the beginning and end of the test period, the detector unit shall respond to a concentration (C) of 1.2 $\pm$ 0.2 gamma per liter of methanesulfonyl fluoride (MSF) (see 6.6), within a time-to-alarm (T) in minutes (min) such that the CT is no more than 0.80 gamma-minutes per liter when tested as specified in 4.4.4.3.3. The cell sensitivity for this test shall be as specified in 3.3.4.3.

3.3.4 Air filter (Drawing C5-15-5307).

3.3.4.1 Pressure drop across the filters. The pressure drop across the filters housed in the air filter shall be between 1.0 and 3.75 inches (2.54 and 9.53 cm) of water at an airflow rate through the filters of 1.1 $\pm$ 0.1 liters per minute when tested as specified in 4.4.4.4.1.

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3.3.4.2 Leakage. Air leakage through the interface seal between the air filter and air filter insert shall not exceed 3.0 cubic centimeters per 15 seconds at an air pressure difference across the interface seal of 45.0 ± 5.0 inches of water when tested as specified in 4.4.4.4.2.

3.3.4.3 Filter conversion of O,S, diethyl methylphosphonothiolate (OSDMP). The air filter when tested under the conditions and parameters listed in table II shall affect a conversion of OSDMP which shall produce an electrical output from the detector cell of at least 15.0 millivolts above background with a maximum CT of 0.30 gamma-minutes per liter when tested in accordance with 4.4.4.4.3.

TABLE II. Conditions and parameters

Condition	Parameter
Filter conditioning	*
OSDMP concentration	0.30 ± .10 gamma/l
Flowrate	1.1 l/min
Temperature	92.5° ± 2.5°F
**Cell sensitivity	21 to 36 mv

*Prior to testing, the air filter shall be aerated for at least 15 minutes but not more than 2.0 hours.

**Cells used for the filter conversion test shall initially conform to MIL-D-51388 or EA-D-1106 and shall be requalified to the noise and background requirement of 3.5.1 and sensitivity requirement of 3.5.2 of MIL-D-51388 or EA-D-1106 once each week prior to use in testing. Noise and background shall be recorded for a period of 10 and 2 minutes respectively. After cell has been in operation for at least 10 minutes, cell background shall not exceed 30 mv during any 12 hour continuous test period.

3.3.4.4 Moisture content. The moisture content of the air filter shall be from 1.0 milligram (mg) to 4.0 mg of water when tested as specified in 4.4.4.4.4.

3.3.5 Low temperature storage. After being stored at -60° ± 5°F (-51.11 ± 2.78°C) for at least 14 days, and then at 80° ± 5°F (26.67 ± 2.78°C) for a minimum of 12 hours as specified in 4.3.3.1, the refill kit shall satisfy the following:

(a) Each chemical reservoir of the refill kit shall contain no less than 90 milliliters (ml) of solution.

(b) The IBA in the refill kit shall show no evidence of discoloration within the packet.

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(c) Each simulant bottle of the refill kit shall contain no less than 4.5 ml of solution.

(d) Each air filter of the refill kit shall be capable of satisfying the conversion requirements of 3.3.4.3.

(e) Each chemical reservoir of the refill kit shall be capable of satisfying the use test requirements of 3.3.3.3.

(f) The concentration of benzenesulfonyl chloride in the simulant solution bottles of the refill kit shall be as specified in 3.3.2.

3.3.6 High temperature storage. After being stored at 150° ± 5°F (65.56° ± 2.78°C) for at least 14 days, and then allowed to return to room temperature as specified in 4.3.3.2, the refill kit shall satisfy the following:

(a) Each chemical reservoir of the refill kit shall contain no less than 90 ml of solution.

(b) The IBA in the refill kit shall show no evidence of discoloration within the packet.

(c) Each simulant bottle of the refill kit shall contain no less than 4.5 ml of solution.

(d) Each air filter of the refill kit shall be capable of satisfying the conversion requirements of 3.3.4.3.

(e) Each chemical reservoir of the refill kit shall be capable of satisfying the use test requirements of 3.3.3.3.

(f) The concentration of benzenesulfony chloride in the simulant solution bottles of the refill kit shall be as specified in 3.3.2.

3.3.7 Cyclic temperature storage. After being stored for 14 days in a cyclic temperature environment varying between temperatures of -60° ± 5°F (-51.11 ± 2.78°C) and 150° ± 5°F (26.67 ± 2.78°C) at a rate of 1.0 ± 0.5 cycle per 24 hours with a maximum time between temperature extremes of 4.0 hours, and then stored at 80° ± 5°F for 12 hours as specified in 4.3.3.3, the refill kit shall satisfy the following:

(a) Each chemical reservoir of the refill kit shall contain no less than 90 ml of solution.

(b) The IBA in the refill kit shall show no evidence of discoloration within the packet.

(c) Each simulant bottle of the refill kit shall contain no less than 4.5 ml of solution.

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(d) Each chemical reservoir of the refill kit shall be capable of satisfying the use test requirements of 3.3.3.3.

(e) The concentration of benzenesulfony chloride in the simulant solution bottles of the refill kit shall be as specified in 3.3.2.

3.3.8 Transportation vibration. After being subjected to the vibration requirements of MIL-STD-810, method 514, procedure XI, part 2 as specified in 4.3.3.4, the refill kit shall satisfy the following:

(a) Each chemical reservoir of the refill kit shall contain no less than 90 ml of solution.

(b) The IBA in the refill kit shall show no evidence of discoloration within the packet.

(c) Each simulant bottle of the refill kit shall contain no less than 4.5 ml of solution.

(d) Each air filter of the refill kit shall be capable of satisfying the conversion requirements of 3.3.4.3.

(e) The moisture content of the air filters shall satisfy the requirements of 3.3.4.4.

(f) The kit shall have no physical deterioration, loose or broken parts.

3.4 Preproduction. Prior to the start of regular production a preproduction sample of refill kits shall be produced in accordance with this specification for examination and test (see 4.3).

3.5 Workmanship. The refill kit, and all components and subassemblies thereof, shall be free from damage and evidence of leakage. Dusting of the conversion filter within the packaged air filter shall not be a cause for rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction.

4.3.1 Sample. A preproduction sample of seven refill kits shall be produced using the same methods, materials, equipment, and processes as will be used in regular production. Components used in the production of the sample of refill kits shall have been examined and tested in accordance with 4.4.

4.3.2 Inspection.

4.3.2.1 For examination and environmental test. The preproduction sample of refill kits shall be examined in accordance with the classification of defects 4.4.3.2(a) and then divided into groups and tested as specified in table III.

4.3.3 Tests. Tests shall be conducted as follows:

4.3.3.1 Low temperature storage. Examine each refill kit and its contents to establish the pretest condition of the refill kit. Do not disturb the closures of the liquid containers. Reassemble the refill kit. Do not band, package, or pack the refill kit. Place the specified quantity (see table III) of refill kits in a test chamber already at the temperature required by 3.3.5. One-half of the sample of refill kits must be placed in the test chamber with the closures of the liquid containers down. Maintain the test chamber at the temperature required by 3.3.5 for at least 14 days. Then, increase the temperature of the test chamber to 80° ± 5°F (26.67 ± 2.78°C). Maintain the test chamber at this temperature at least 12 hours. Remove the sample of refill kits from the test chamber. Examine the liquid level of each chemical reservoir. Measure the contents of reservoirs suspected of containing less than the quantity solution required by 3.3.5. Examine each IBA packet for discoloration. Measure the contents of the simulant solution bottles. Subject the air filter sample required by table III to the conversion test required by 3.3.4.3. Subject the simulant bottle sample required by table III to the benzenesulfonyl chloride concentration test required by 3.3.2. Evaluate the findings of the post storage examinations and tests against the requirements of 3.3.5.

4.3.3.2 High temperature storage. Examine each refill kit and its contents to establish the pretest condition of the refill kit. Do not disturb the closures of the liquid containers. Reassemble the refill kit. Do not band, package, or pack the refill kit. Place the specified quantity (see table III) of refill kits in a test chamber already at the temperature required by 3.3.6.

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One-half of the sample of refill kits must be placed in the test chamber with the closures of the liquid containers down. Maintain the test chamber at the temperature required by 3.3.6 for at least 14 days. Then, remove the refill kit sample from the test chamber. Open the kits, and allow the kit contents to cool to room temperature. Examine the liquid level of each chemical reservoir. Measure the contents of reservoirs suspected of containing less than the quantity of solution required by 3.3.6. Examine each IBA packet for discoloration. Measure the contents of the simulant solution bottles. Subject the air filter sample required by table III to the conversion test required by 3.3.4.3. Subject the chemical reservoir sample required by table III to the test required by 3.3.3.3. Subject the simulant bottle sample required by table III to the benzenesulfonyl chloride concentration test required by 3.3.2. Evaluate the findings of the post storage examinations and test against the requirements of 3.3.6.

4.3.3.3 Cyclic temperature storage. Examine each refill kit and its contents to establish the pretest condition of the refill kit. Do not disturb the closures of the liquid containers. Reassemble the refill kit. Do not band, package, or pack the refill kit. Place the specified quantity (see table III) of refill kits in the test chamber. One-half of the sample of refill kits must be placed in the test chamber with the closures of the liquid containers down. Start the test chamber cycling within the limits required by 3.3.7, and continue the cycling for at least 14 days. Then, stop the cycling of the chamber, and set the chamber controls to produce a temperature of 80° $\pm$ 5°F (26.67 $\pm$ 2.78°C). Maintain this temperature at least 12 hours. Remove the sample of refill kits from the test chamber. Examine the liquid level of each chemical reservoir. Measure the contents of reservoirs suspected of containing less than the quantity of solution required by 3.3.7. Examine each IBA packet for discoloration. Measure the contents of the simulant bottles. Subject the chemical reservoir sample required by table III to the use test required by 3.3.3.3. Subject the simulant bottle sample required by table III to the benzenesulfonyl chloride concentration test required by 3.3.2. Evaluate the findings of the post storage examinations and test against the requirements of 3.3.7.

4.3.3.4 Transportation vibration. Examine each refill kit and its contents to establish the pretest condition of the refill kit. Do not disturb the closures of the liquid containers. Reassemble and band the refill kit, and then package and pack it as described in section 5. Subject the specified quantity (see table III) of refill kits to the vibration test described in MIL-STD-810, method 514, procedure XI, part 2. After completion of the prescribed test, examine the liquid level of each chemical reservoir. Measure the contents of the reservoirs suspected of containing less than the quantity of solution required by 3.3.8. Examine each IBA packet for discoloration. Measure the contents of the simulant bottles. Subject the air filter sample required by table III to the conversion and moisture tests required by 3.3.4.3 and 3.3.4.4. Evaluate the findings of the post vibration examinations and test against the requirements of 3.3.8. Examine for physical deterioration, loose or broken parts.

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TABLE III. Preproduction sample inspection

Quantity	Component	Requirement	Component sample size	AQL(%) or acc/rej no.	Test paragraph
2 Refill kits	IBA packet	Color 3.3.5(b)	Each chemical reservoir	Maximum 1 per refill kit	4.3.3.1
	Chemical reservoir	Solution content 3.3.5(a)	Each chemical reservoir	Maximum 1 per refill kit	4.3.3.1
		Use test 3.3.5(e)	One chemical reservoir	Acc 0, rej 1	4.3.3.1
	Simulant solution bottle	Solution content 3.3.5(c)	Each simulant bottle	Acc 0, rej 1	4.3.3.1
		Benzenesulfonyl chloride concentration 3.3.5(f)	One simulant bottle	Acc 0, rej 1	4.3.3.1
	Air filter	OSDMP conversion 3.3.5(d)	55	Acc 2, rej 3	4.3.3.1
2 Refill kits	IBA packet	Color 3.3.6(b)	Each chemical reservoir	Maximum 1 per refill kit	4.3.3.2
	Chemical reservoir	Solution content 3.3.6(a)	Each chemical reservoir	Maximum 1 per refill kit	4.3.3.2
		Use test 3.3.6(e)	One chemical reservoir	Acc 0, rej 1	4.3.3.2
	Simulant solution bottle	Solution content 3.3.6(c)	Each simulant bottle	Acc 0, rej 1	4.3.3.2
		Benzenesulfonyl chloride concentration 3.3.6(f)	One simulant bottle	Acc 0, rej 1	4.3.3.2

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TABLE III. Preproduction sample inspection (Cont'd)

Quantity	Component	Requirement	Component sample size	AQL(%) or acc/rej no.	Test paragraph
	Air filter	OSDMP conversion 3.3.6(d)	55	Acc 9, rej 10	4.3.3.2
2 Refill kits	IBA packet	Color 3.3.7(b)	Each chemical reservoir	Maximum 1 per refill kit	4.3.3.3
	Chemical reservoir	Solution content 3.3.7(a)	Each chemical reservoir	Maximum 1 per refill kit	4.3.3.3
		Use test 3.3.7(d)	One chemical reservoir		
	Simulant solution bottle	Solution content 3.3.7(c)	Each simulant bottle	Acc 0, rej 1	4.3.3.3
		Benzenesulfonyl chloride concentration 3.3.7(e)	One simulant bottle	Acc 0, rej 1	4.3.3.3
1 Refill kit	IBA packet	Color 3.3.8(b)	Each chemical reservoir	Maximum 1 per refill kit	4.3.3.4
	Chemical reservoir	Solution content 3.3.8(a)	Each chemical reservoir	Maximum 1 per refill kit	4.3.3.4
	Simulant solution bottles	Solution content 3.3.8(c)	Each simulant bottle	Acc 0, rej 1	4.3.3.4
	Air filter	OSDMP conversion 3.3.8(d)	55	Acc 2, rej 3	4.3.3.4
		Moisture content	44	Acc 0, rej 1	4.4.4.4.4

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4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with the examination and tests specified in 4.3.3 and 4.4.4 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting.

(a) Refill kits. A lot shall consist of the refill kits offered for acceptance at one time which have been produced by one manufacturer, at one plant, under essentially the same manufacturing conditions, and in essentially the same manufacturing conditions, and in essentially the same time frame. However, not more than one batch (see 6.7) of IBA or one lot of air filters shall be represented in any one lot of finished refill kits.

(b) Air filters. A lot shall consist of the air filters manufactured from the filter material which has been impregnated from one batch of silver nitrate solution.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling of refill kits or components shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. Sampling of components shall be conducted in accordance with table IV.

4.4.3 Inspection procedure.

4.4.3.1 For examination and tests. Sample refill kits and components shall be examined and tested in accordance with the classification of defects (4.4.3.2), and MIL-STD-105, and table IV.

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TABLE IV. Component sampling and inspection

component	Requirement	Military standard or special sample size	AQL(%) or acc/rej no.	Test para
IBA	Melting point, chemical composition, water content, particle size, color 3.3.1	One sample (10.0 ± 0.5 grams) from each batch	Acc 0, rej 1	4.4.4.1
Bottled simulant solution	Benzenesulfonyl chloride concentration 3.3.2	1 bottle per 1,000 or 1 bottle per batch (see 6.7) of simulant solution, whichever quantity of simulant solution is smaller.	Acc 0, rej 1	4.4.4.2
Chemical reservoir	Powder packet orientation contamination 3.3.3.1	MIL-STD-105	2.5	4.4.4.3.1
	Cutter-plunger operation 3.3.3.2	Two from sample taken for 3.3.3.1	Acc 0, rej 1	4.4.4.3.2
	Use test 3.3.3.3	Two from sample taken for 3.3.3.1	Acc 0, rej 1	4.4.4.3.3
Air filter	Pressure drop 3.3.4.1	MIL-STD-105	2.5	4.4.4.4.1
	Leakage 3.3.4.2	MIL-STD-105	2.5	4.4.4.4.2
	OSDMP conversion 3.3.4.3	MIL-STD-105	2.5	4.4.4.4.3
	Moisture 3.3.4.4	MIL-STD-105	*2.5	4.4.4.4.4

*Inspection level S4.

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4.4.3.2 Classification of defects.

(a) Refill kit (Drawing D5-15-4700).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Component missing, incorrect or incorrectly assembled
102	Printed instructions missing, incorrect, or illegible
103	Damage: kit or component packages ripped, torn, or punctured
104	Carrying handle broken or incorrectly mounted

(b) Package assembly (Drawing C5-15-5309).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Air filter package: inner or outer package missing, incorrect, or incorrectly assembled
102	Air filter package: inner or outer package ripped, torn, or punctured
103	Air filter component or filter missing, incorrect, or incorrectly assembled
104	Air filter (other than handle) or insert cracked, gouged or warped
105	Filters ripped, torn, punctured, or cracked

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